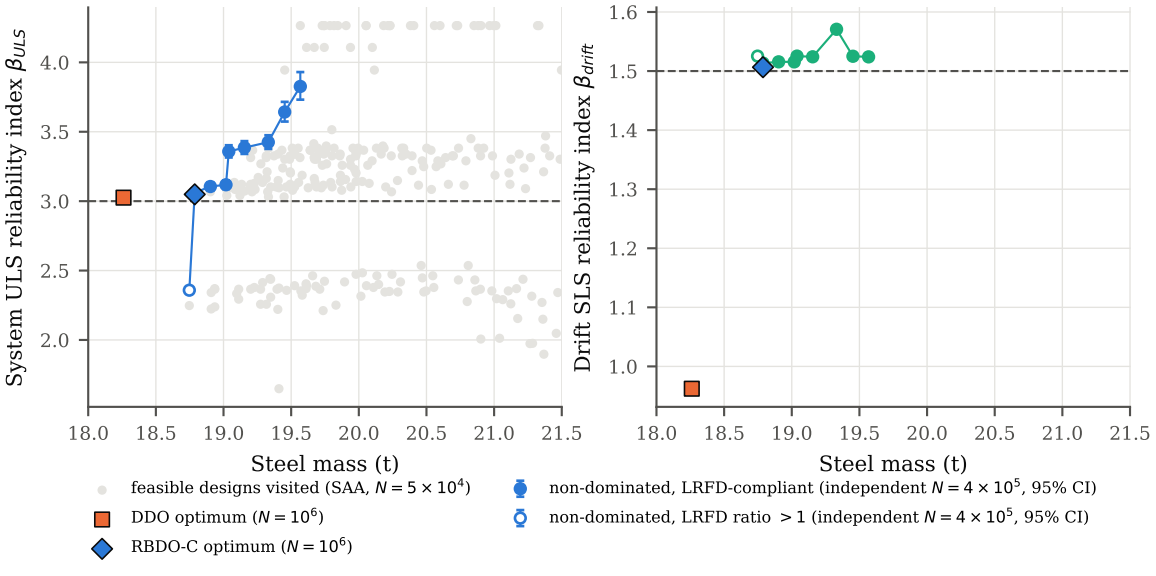
(a) mass vs. ultimate reliability
(b) drift reliability of the same designs
System ULS reliability index β_{ULS}
Drift SLS reliability index β_{drift}
Steel mass (t)
Steel mass (t)
feasible designs visited (SAA, $N = 5 \times 10^4$)
DDO optimum ($N = 10^6$)
RBDO-C optimum ($N = 10^6$)
non-dominated, LRFD-compliant (independent $N = 4 \times 10^5$, 95% CI)
non-dominated, LRFD ratio > 1 (independent $N = 4 \times 10^5$, 95% CI)